

Introduction

Abstract:

- Deep reinforcement learning (DRL) can learn fast control policies for complex systems, but its performance often degrades when deployment conditions differ from training.
- Bounded extremum seeking (ES) provides model-independent online adaptation with bounded control updates, making it suitable for noisy, unknown, and time-varying systems.
- We combine DRL and bounded ES into a hybrid ES-DRL controller: DRL provides fast task execution, while ES adapts online when distribution shift or changing dynamics reduce RL performance.
- In robotic manipulation, ES-DRL improves robustness for Fetch pushing and pick-and-place tasks under time-varying goals and spatially varying friction, where RL alone loses contact or fails to track the target.
- The same framework also improves accelerator tuning by adapting quadrupole magnet settings under time-varying lattice perturbations.

DRL mode:

- Uses the trained actor mode.
- Provides fast, directed motion.
- Handles task entry, reaching, and contact.
- **DDPG training objective:** $J(\mu_\theta) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t]$
- Actor: $a_t = \mu_\theta(s_t)$; Critic: $Q(s_t, a_t)$.

ES mode:

- Uses scalar performance feedback.
- Adapts online without requiring a system model.
- Provide robustness when dynamics shift and contact behavior changes.

$$u_i = \sqrt{\alpha\omega_i} \cos(\omega_i + ky)$$

Time-varying systems

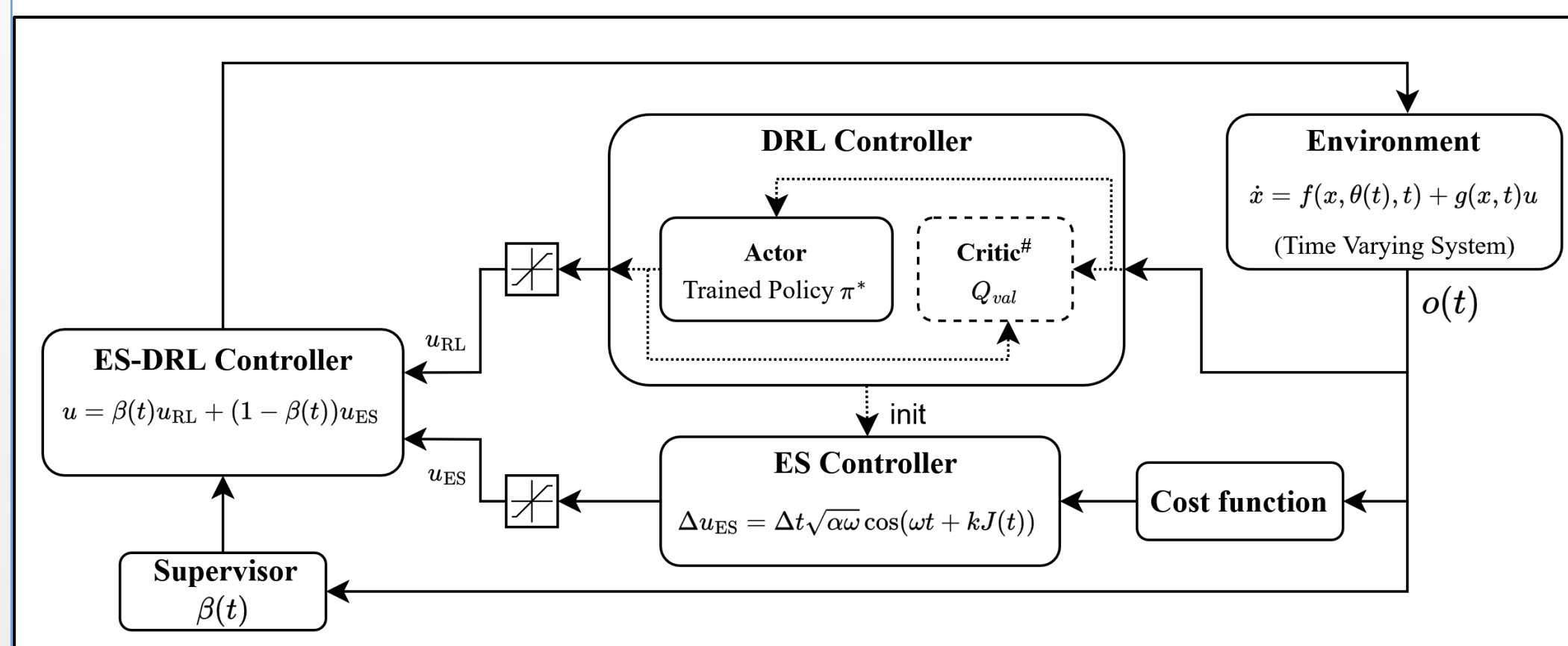
System	State	Action
LEBT accelerator tuning	4x4000	22 (Quadrupole magnet strengths)
Fetch Robot	28	4 (dx, dy, dz, gripper position)

ES-DRL Controller

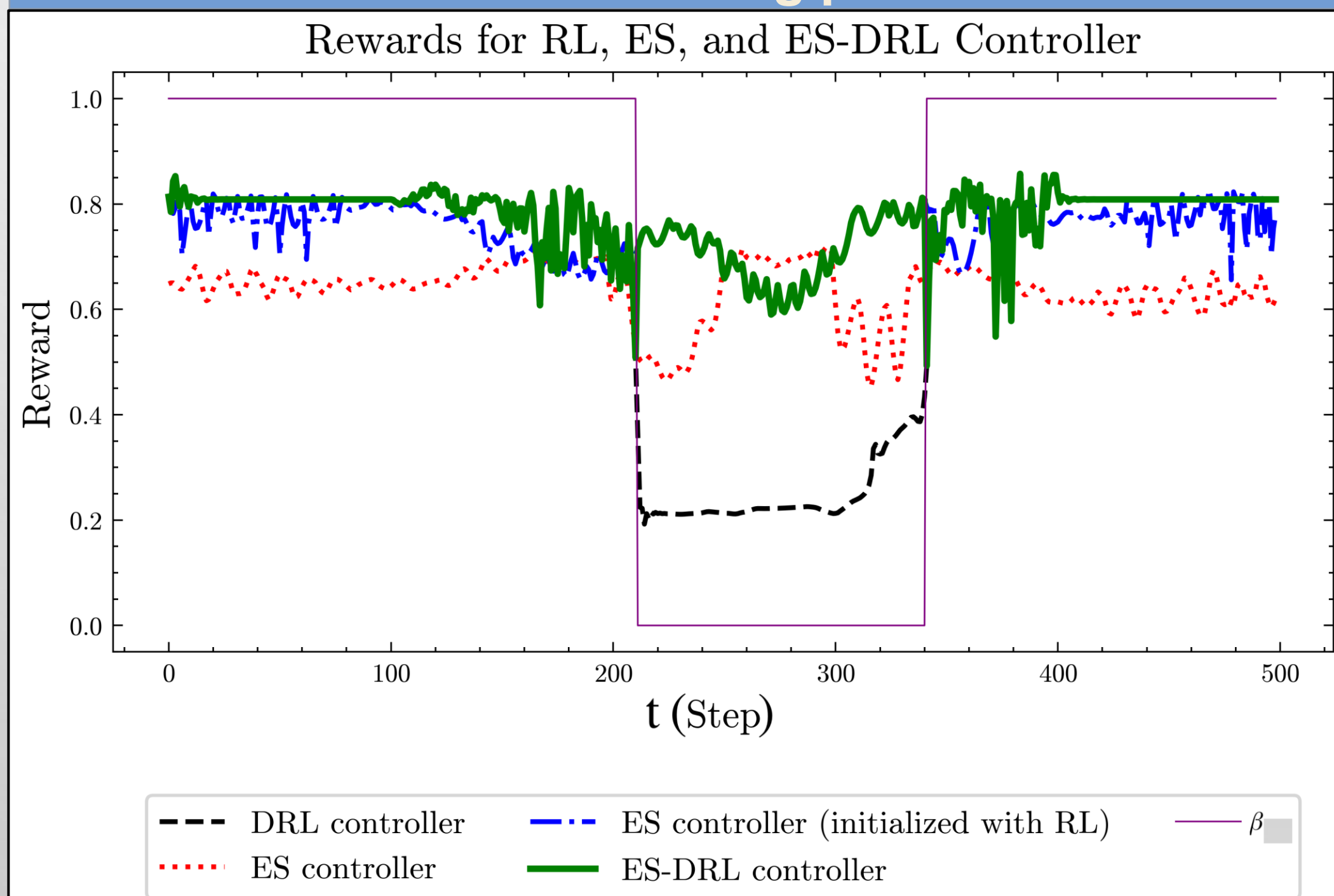
The proposed controller combines a learned DRL policy with a bounded ES adaptive controller:

$$u = \beta u_{RL} + (1 - \beta) u_{ES}$$

- The supervisor β selects the control mode based on task state or safety conditions.
- The ES-controller is warm started from the DRL action, reducing transient and allowing adaptation to begin from a useful control direction.

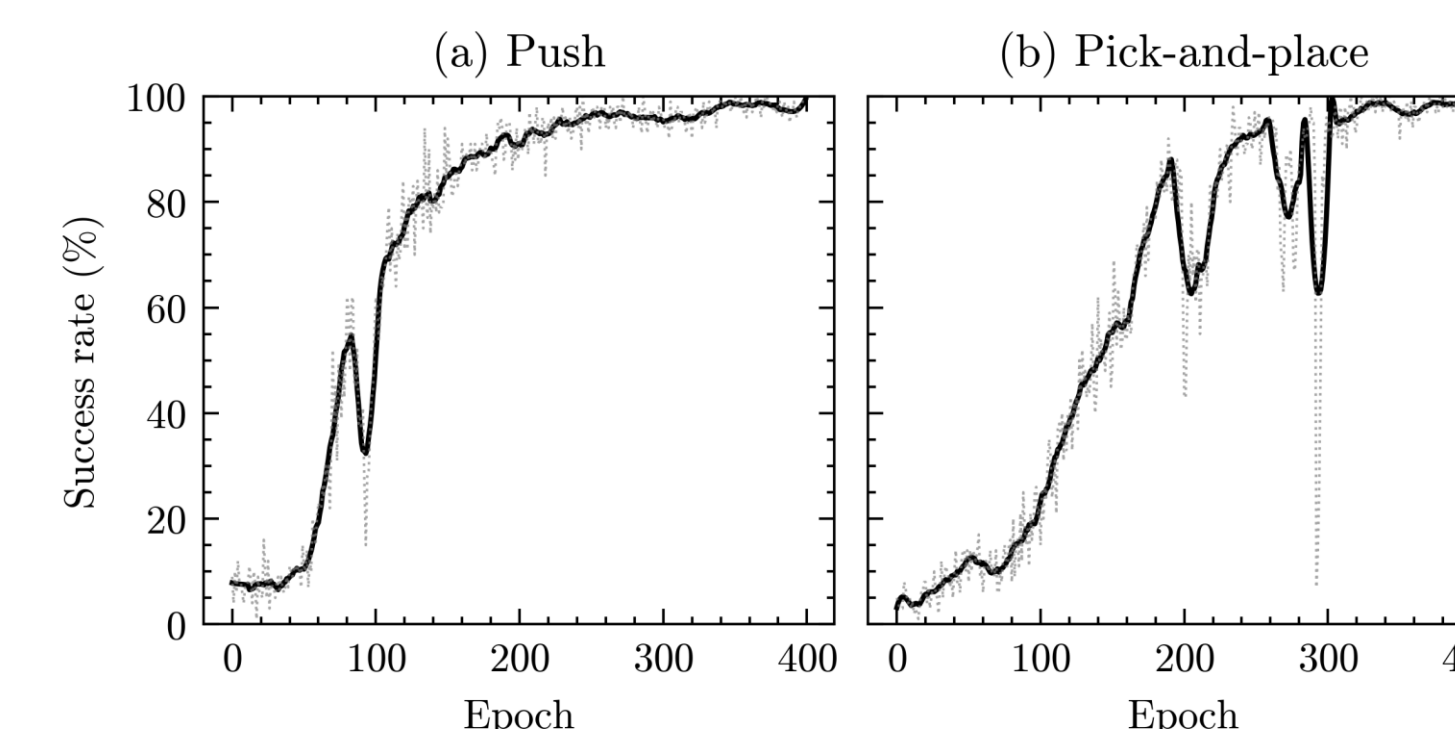


Accelerator tuning problem



Robotic manipulation

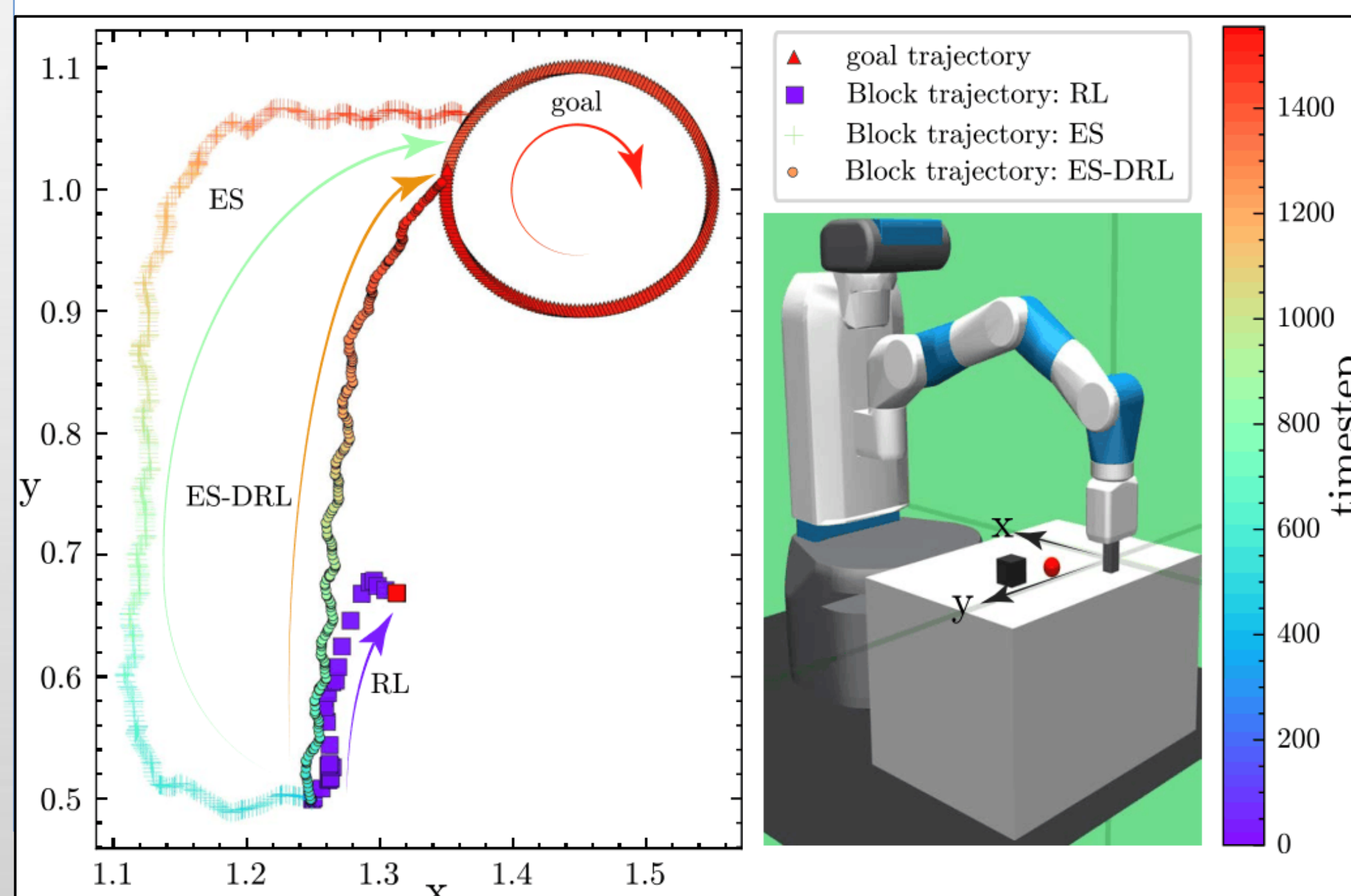
- DDPG policies are trained on Fetch push and pick-and-place tasks.



- State includes robot pose, object pose, object velocity, gripper state, and desired goal.
- Action is a 4D command: Cartesian end-effector displacement plus gripper command.
- Reward encourages the robot to reach the object and move the object toward the goal.

Deployment shifts:

- Time-varying goal
- Spatially varying frictional patches.



Supervisor β

Accelerator tuning problem:

$$\beta = \begin{cases} 1, & \text{if } \bar{x}(t) \text{ or } \bar{y}(t) < 0.7 r_{max} (RL \text{ mode}) \\ 0, & \text{otherwise (ES mode)} \end{cases}$$

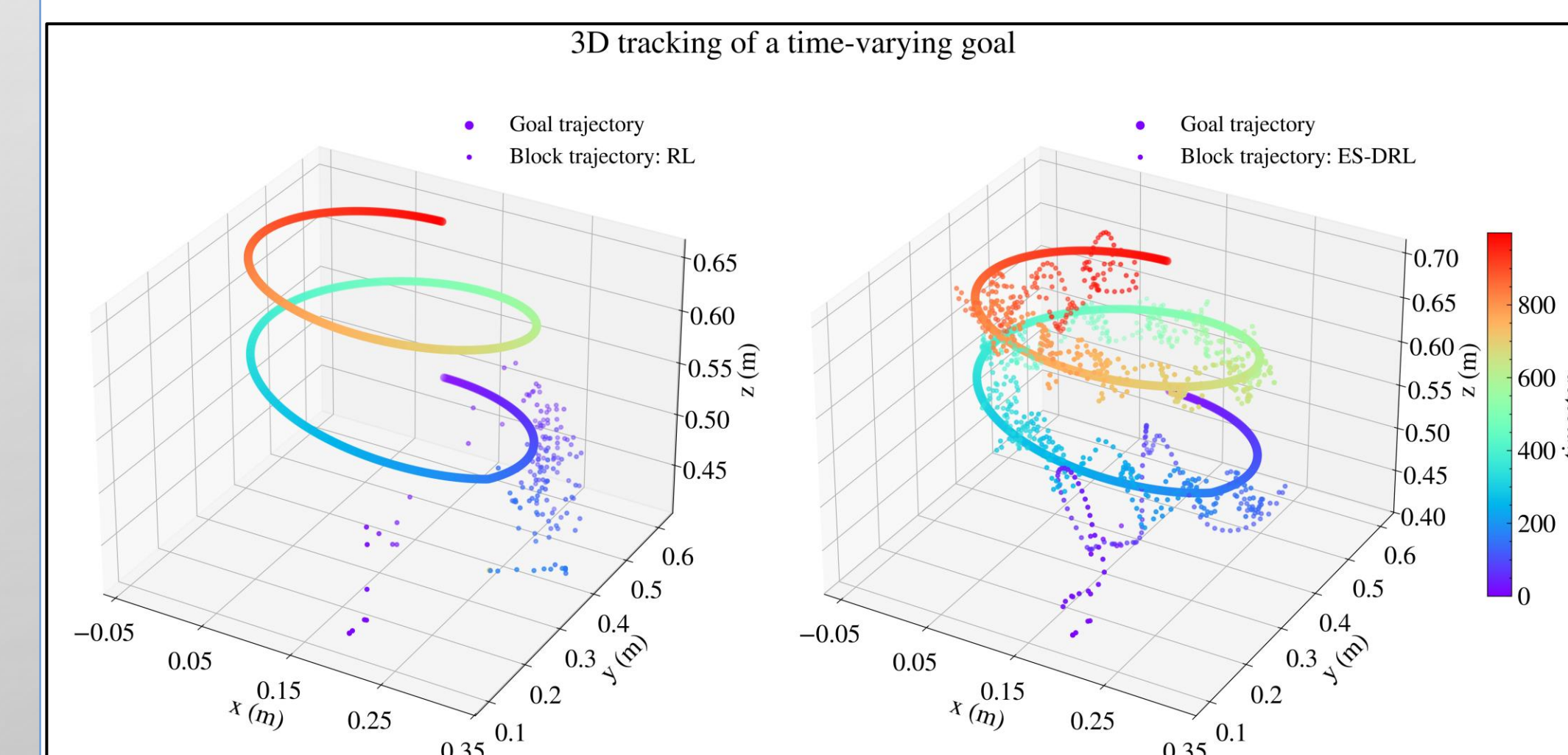
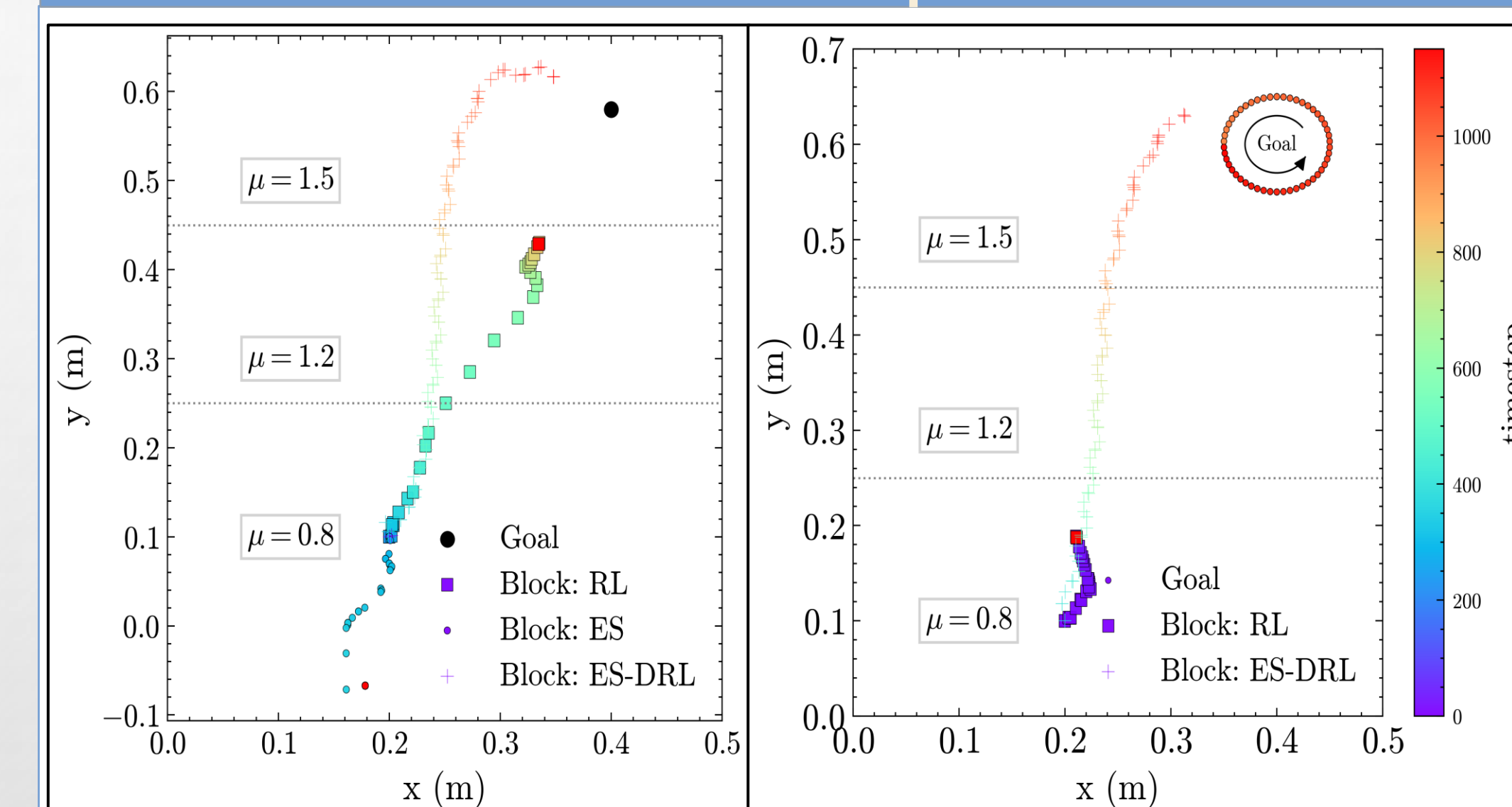
where, $r_{max} = 25.4 \text{ mm}$ denote the beam-pipe radius.

Robotic manipulation problem:

$$\beta = \begin{cases} 1, t < t_c (RL \text{ mode}) \\ 0, t \geq t_c (ES \text{ mode}) \end{cases}$$

where, t_c denotes the time when gripper meets the object.

Results: Robotic Manipulation tasks



Contact

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References

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Acknowledgement

This work was supported by the U.S. Department of Energy (DOE), Office of Science, Office of High Energy Physics contract number 9233218CNA000001 and the Los Alamos National Laboratory LDRD Program Directed Research (DR) project 20220074DR.